

# NEURALPLANE: STRUCTURED 3D RECONSTRUCTION IN PLANAR PRIMITIVES WITH NEURAL FIELDS

Hanqiao Ye, Yuzhou Liu, Yangdong Liu, Shuhan Shen

School of Artificial Intelligence, University of Chinese Academy of Sciences  
Institute of Automation, Chinese Academy of Sciences

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We aim for structured plane-based 3D reconstruction

- We target clean, semantic planar primitives from casual multi-view capture.
- We find cross-view inconsistencies and shape–radiance ambiguity are the core challenges.
- We observe RANSAC pipelines miss recall; neural surfaces lack structure and semantics.

We fuse 2D plane cues into a unified 3D neural plane field

- We distill noisy 2D plane observations into a 3D neural representation.
- We jointly learn geometry and semantics without plane annotations.
- We extract crisp, semantically aligned planar primitives.

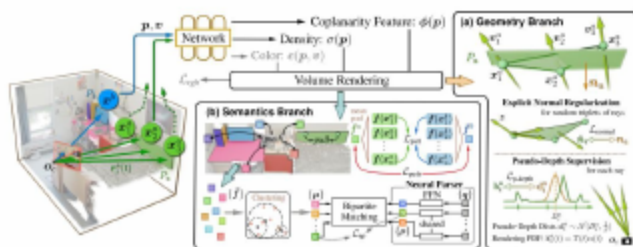


Figure: Plane-guided neural learning overview: geometry and semantics are trained jointly.

We generate 2D plane observations and initialize Local Planar Primitives

- We cluster monocular normals and intersect with SAM over-segmentation for smooth, semantic masks.
- We initialize plane normals by segment averages and offsets from weighted SfM keypoint distances.
- These LPPs provide strong constraints that guide learning.

## We guide geometry with normal consistency and pseudo-depth

- We enforce explicit normal consistency from termination-point triplets to reduce shape–radiance ambiguity.
- We align rendered depth distributions to plane-derived pseudo-depth via KL and jointly refine plane parameters.

## We learn coplanarity features and semantic prototypes

- We train a Neural Coplanarity Field with contrastive pull/push under geometric gating.
- We learn semantic prototypes online (Neural Parser) instead of brittle post-hoc clustering.
- We group termination points by learned prototypes and then fit global planes.



Figure: Neural Coplanarity Field visualization: view-consistent coplanar regions emerge.

## We evaluate on ScanNetv2 and ScanNet++

- We show state-of-the-art geometry and segmentation versus strong baselines.
- We achieve balanced precision/recall and large gains in segmentation quality.
- We train per scene in minutes, enabling practical deployment.

## We reconstruct crisp planes with coherent semantics

- We show clean planar structures and sharp edges across scenes.
- We maintain semantic consistency across views and through occlusions.

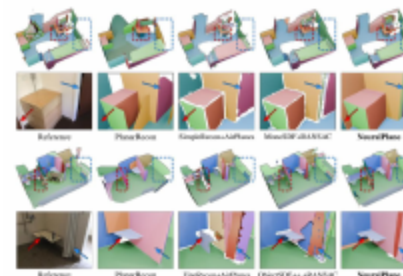


Figure: Qualitative results on ScanNetv2: crisp planes and aligned semantics.

- How can we better handle non-planar and mixed-planarity regions without over-planarizing?
- Can we reduce reliance on monocular/SfM priors and remain robust in the wild?
- How do we scale to large, cluttered, or outdoor scenes while preserving speed?
- Can we integrate edges/intersections and support online SLAM in real time?